

CSE 4849 / CSE 4885
Human Computer Interaction (HCI)
Lecture 3: The Interaction

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The Interaction

- Interaction models
 - Translations between user and system
- Ergonomics
 - Physical characteristics of interaction
- Interaction styles
 - The nature of user/system dialog
- Context
 - Social, organizational, motivational

What is interaction?

- Interaction is the communication between user and the system.

user ↔ system

- Communication happens through the interface.
- Interface must therefore effectively translate between them to allow the interaction to be successful.
- Model of interaction can help us to understand the interaction details and identify the possible difficulties.

Models of interaction

Terms of interaction

Norman model

Interaction framework

Some terms of interaction

Domain	The area of work under study. Example, Graphic design.
Goal	What you want to achieve. Example, Create a solid red triangle.
Task	How you go about doing it in terms of operations or actions. Example, Select fill tool, click over triangle.
Intention	Intention is the specific action required to meet the goal.
Task analysis	Involves the identification of problem space in terms of domain, goals, tasks and intentions.
System language	Core language describes computational attributes of the domain relevant to system state.
Task language	Describes psychological attributes of the domain related to user state.

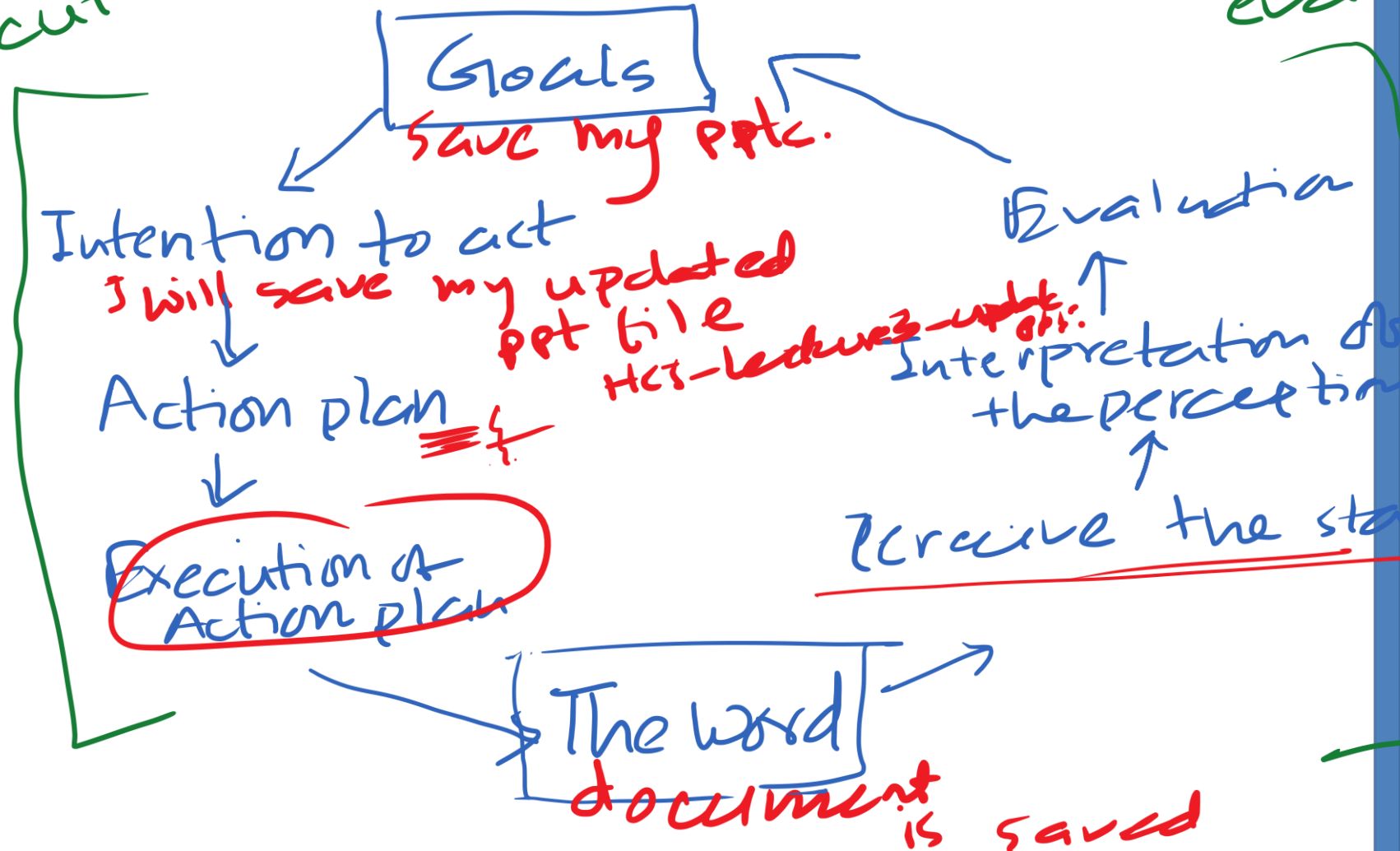
Donald Norman's model of Interaction

- Seven stages
 - User establishes the goal
 - Formulates intention
 - Specifies actions at interface
 - Executes action
 - Perceives system state
 - Interprets system state
 - Evaluates system state with respect to goal
- Norman's model concentrates on user's view of the interface

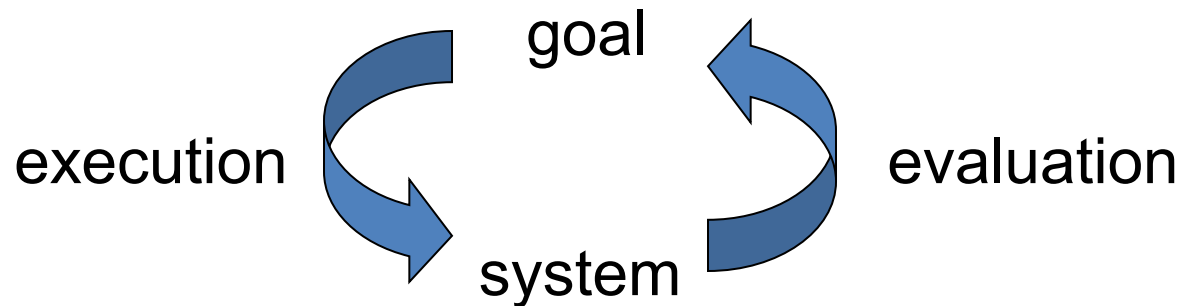
Norman Model of task performance

Execution

eva

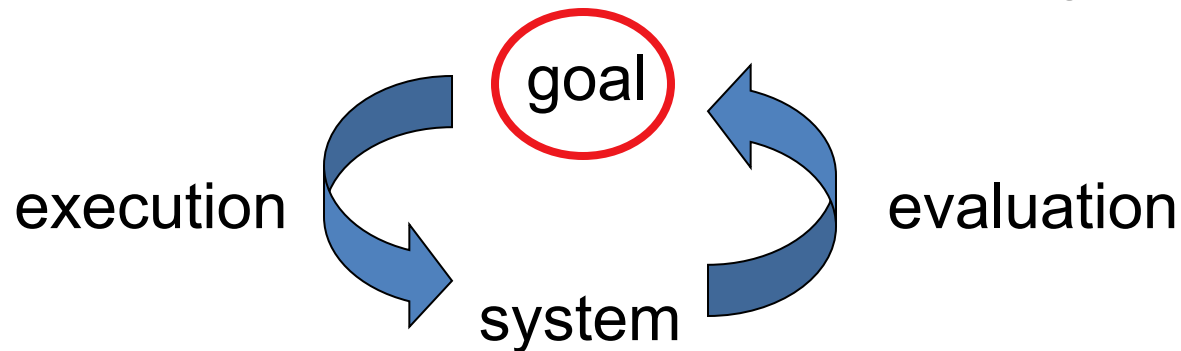


execution/evaluation loop



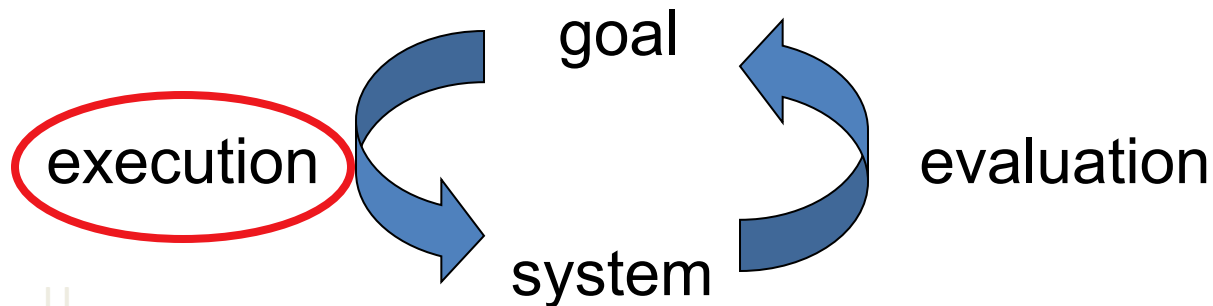
1. User establishes the goal
2. Formulates intention
3. Specifies actions at interface
4. Executes action
5. Perceives system state
6. Interprets system state
7. Evaluates system state with respect to goal

execution/evaluation loop



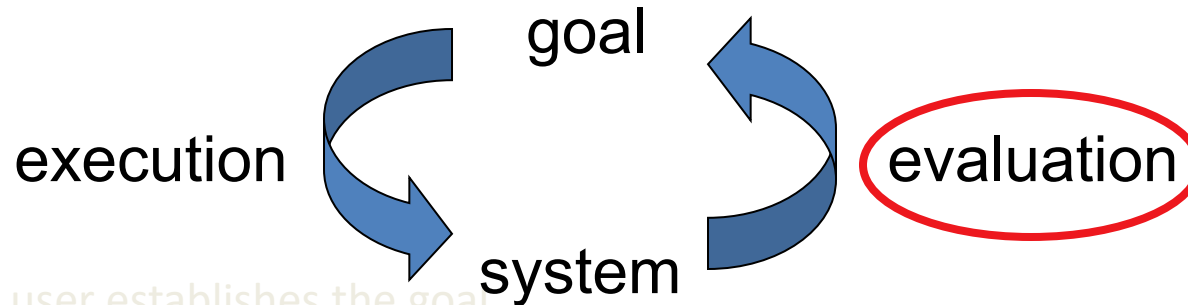
1. User establishes the goal
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execution/evaluation loop



1. User
2. User establishes the goal
3. Formulates intention
4. Specifies actions at interface
5. Executes action
6. Perceives system state
7. Interprets system state
8. Evaluates system state with respect to goal

execution/evaluation loop



1. user establishes the goal
2. formulates intention
3. specifies actions at interface
4. executes action
5. Perceives system state
6. Interprets system state
7. Evaluates system state with respect to goal

Using Norman's model

Some systems are harder to use than others

Gulf of Execution

user's formulation of actions to reach the goal

≠ actions allowed by the system

Gulf of Evaluation

Physical presentation of the system state

≠ the expectation of the user



Human error - slips and mistakes

slip



understand system and goal



correct formulation of action



incorrect action

mistake



may not even have right goal!

Fixing things?

slip – better interface design

mistake – better understanding of system

Abowd and Beale framework

extension of Norman...

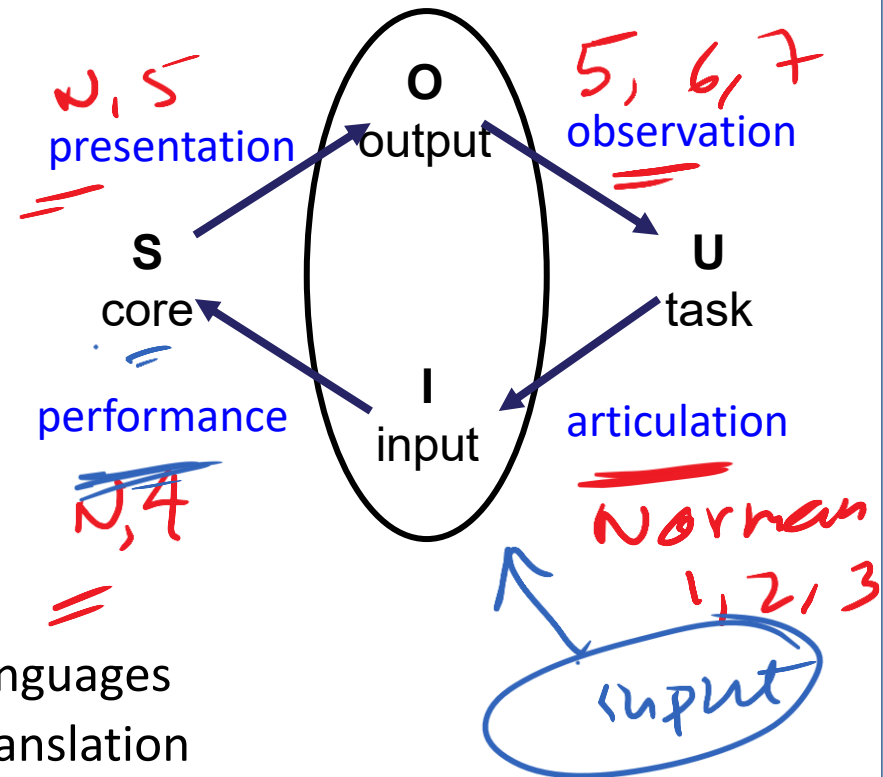
their interaction framework has 4 parts

- user
 - input
 - system
 - Output
- :)

each has its own unique language

interaction $\Rightarrow$ translation between languages

problems in interaction = problems in translation



1. User establishes the goal
2. Formulates intention
3. Specifies actions at interface
4. Executes action
5. Perceives system state
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A framework of HCI concerning interaction issues

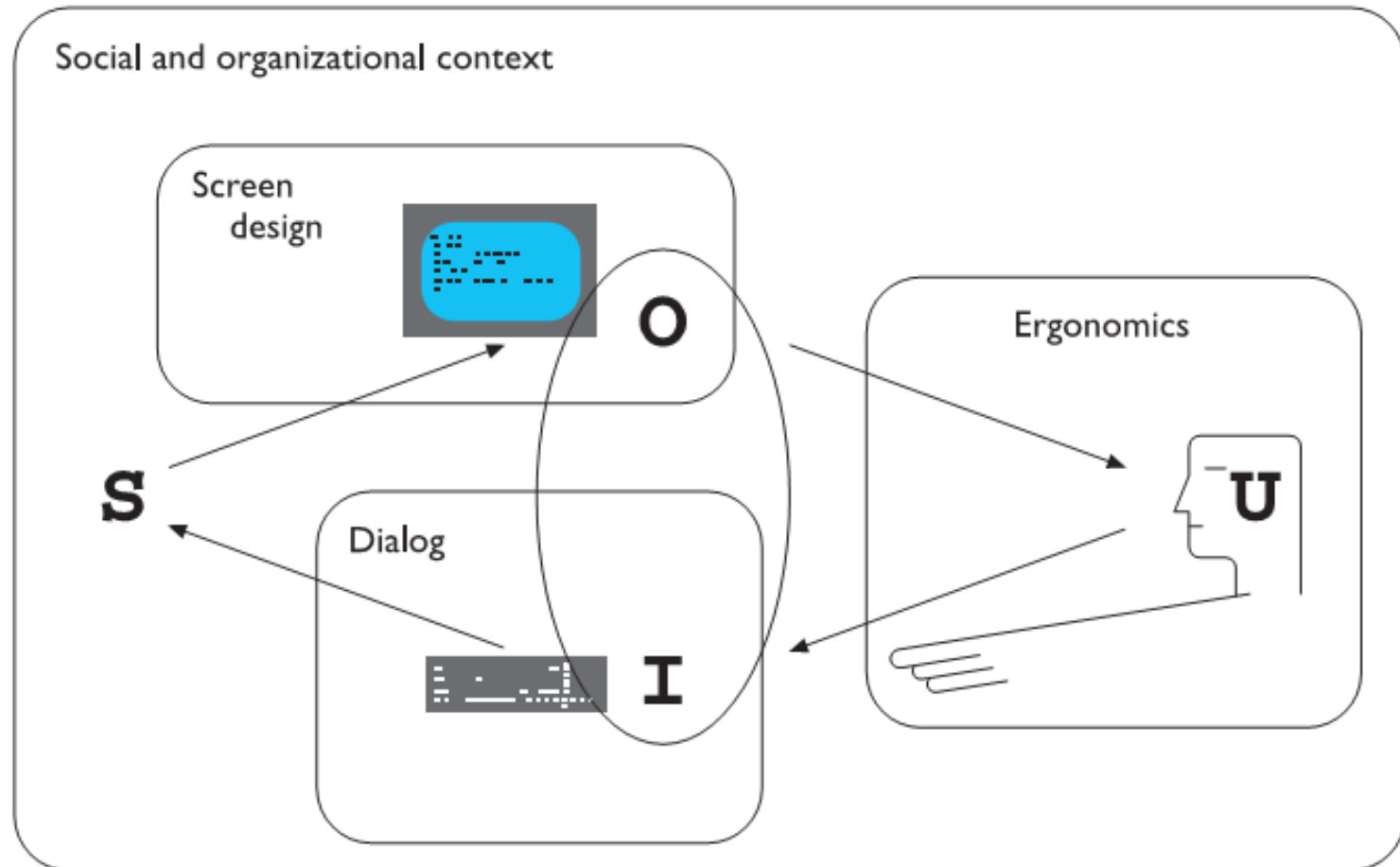


Figure 3.3 A framework for human–computer interaction. Adapted from ACM SIGCHI Curriculum Development Group [9]

Ergonomics

physical aspects of interfaces

industrial interfaces

Ergonomics

- Study of the physical characteristics of interaction
 - How the controls are designed
 - Physical environment in which the interaction takes place
 - Layout and physical qualities of the screen
- Known as human factors that also touch upon human psychology and system constraints.
- Ergonomics good at defining standards and guidelines for constraining the way we design certain aspects of systems

Ergonomics - examples

1. Arrangement of controls and displays

e.g. controls grouped according to

- **Function:** functional controls and displays are organized so that those that are functionally related are placed together.
- **Frequency of use:** controls and displays are organized according to how frequently they are used, with the most commonly used controls being the most easily accessible
- **Sequentially:** controls and displays are organized to reflect the order of their use in a typical interaction.
(aviation)

Ergonomics – examples...

2. Surrounding environment

e.g. Seating arrangements adaptable to cope with all sizes of user

- Where will the system be used?
- By whom will it be used?
- Will users be sitting, standing or moving about?

3. Health issues

e.g. physical position, environmental conditions (temperature, humidity), lighting, noise, time

4. Use of colour

e.g.

- Blue should not be used to display critical information
- Colour should corresponds to common conventions and user expectations. use of red for warning, green for okay, awareness of colour-blindness etc.

Office workstation ergonomics

Head

Head back,
chin tucked,
Ears, shoulder,
hips aligned.

Neck

Use headphones.
Do not cradle
phone between
head and
shoulder!

Elbows

At sides - slightly
more than 90
degree bend.

Chair

Fully adjustable with
lumbar support in
small of the back.

Eyes

Level with top
1/3 of screen.
18-24"

Document Holder

Adjacent to and at
same height as
monitor.

Keyboard

Same height as elbow
with wrists slightly bent.
Keystroke gently!

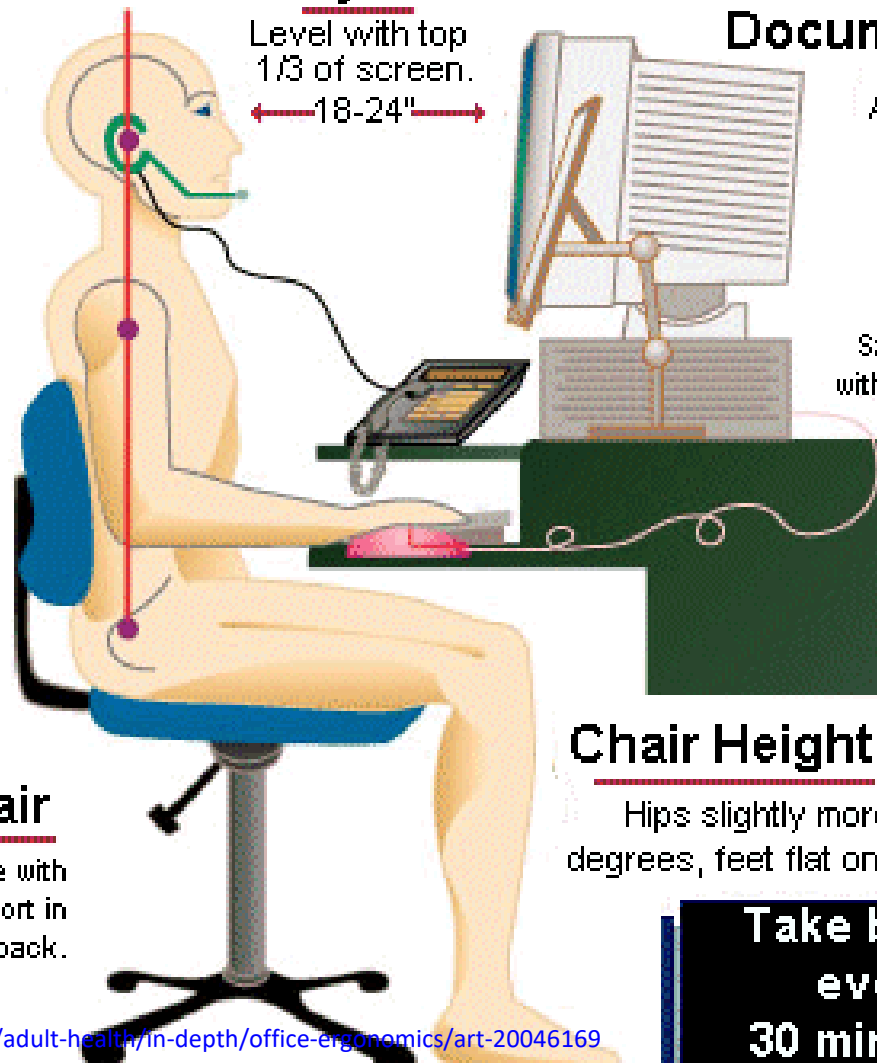
Mouse

Adjacent to and
at same height
as keyboard.

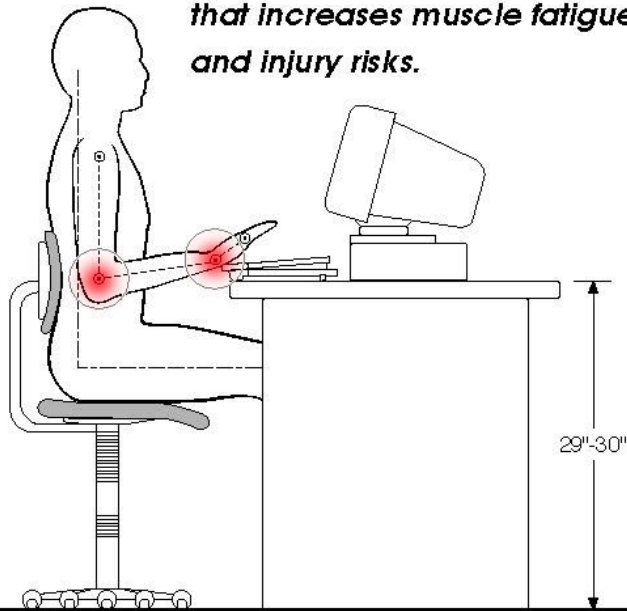
Chair Height

Hips slightly more than 90
degrees, feet flat on the floor

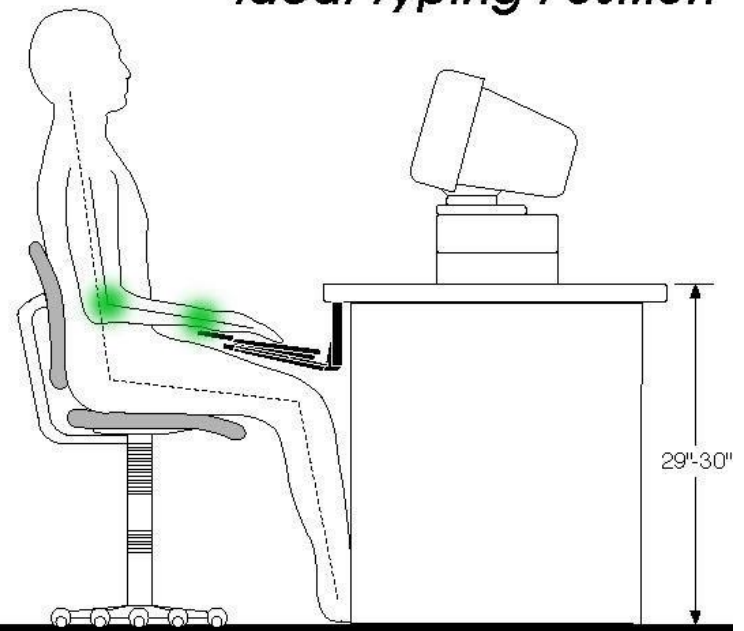
**Take breaks
every
30 minutes!**



*Typical desk top typing posture
that increases muscle fatigue
and injury risks.*



Ideal Typing Position



*Courtesy: Cornell Human Factors and Ergonomics Research Group (CHFERG)
<http://ergo.human.cornell.edu/default.htm>*

9. center monitor and keyboard in front of you

4. sit at arms length from monitor

3. no glare on screen use an antireflective anti-glare

6. use a document holder preferably in-line

7. wrists flat and straight in relation to

forearm 8. arms and elbows relaxed close to body

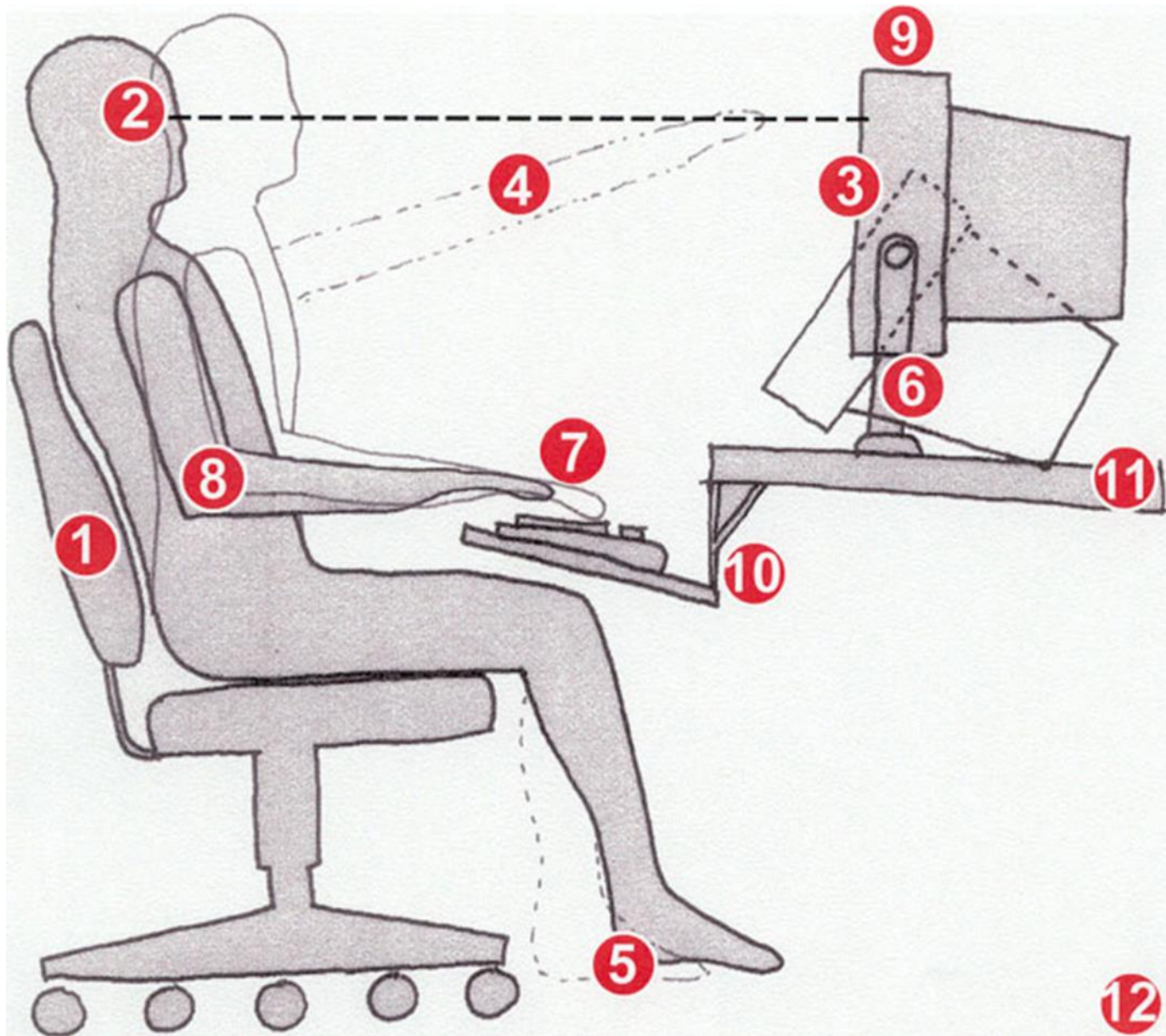
11. use a stable work surface and stable (no bounce) keyboard tray

1. use a good chair with a dynamic chair back and sit back in this

10. use a negative tilt keyboard tray with an upper mouse platform or downward tiltable platform adjacent to keyboard

5. feet on floor or stable footrest

12. take frequent short breaks (microbreaks)



12 tips for an Ergonomic Computer Workstation

1. Use a good chair with a dynamic chair back and sit back in this
2. Top of monitor casing 2-3" (5-8 cm) above eye level
3. No glare on screen, use an optical glass anti-glare filter where needed
4. Sit at arms length from monitor
5. Feet on floor or stable footrest
6. Use a document holder, preferably in-line with the computer screen
7. Wrists flat and straight in relation to forearms to use keyboard/mouse/input device
8. Arms and elbows relaxed close to body
9. Center monitor and keyboard in front of you
10. Use a negative tilt keyboard tray with an upper mouse platform or downward tiltable platform adjacent to keyboard
11. Use a stable work surface and stable (no bounce) keyboard tray
12. Take frequent short breaks (microbreaks)



Interaction styles

Dialogue ... computer and user

Distinct styles of interaction

Common interaction styles

- Command line interface
- Menus
- Natural language
- Question/answer and query dialogue
- Form-fills and spreadsheets
- WIMP
- Point and click
- Three-dimensional interfaces

Context

Interaction affected by social and organizational context

- Other people
 - Desire to impress, competition, fear of failure
- Motivation
 - Fear, allegiance, ambition, self-satisfaction
- Inadequate systems
 - Cause frustration and lack of motivation